

# **Introduction to Autonomous Agents**

# Autonomous Agents

- An **autonomous agent** is a system or program that:
- Acts **independently**
- Makes decisions on its own
- Interacts with its environment
- It does **not need constant human control**

An autonomous agent is an intelligent entity that perceives its environment and takes actions independently to achieve specific goals.

- **Examples**
- Chatbots
- Self-driving cars
- Virtual assistants

# Characteristics of Agentic Software

**Agentic software** refers to systems built using autonomous agents.

## Key Characteristics

### ◆ 1. Autonomy

Works without human intervention

### ◆ 2. Reactivity

Responds to changes in environment

### ◆ 3. Proactiveness

Takes initiative (goal-oriented behaviour)

### ◆ 4. Social Ability

Communicates with other agents

### ◆ 5. Learning Ability

Improves performance over time

### ◆ 6. Adaptability

Adjusts to new situations

# Differences between Traditional AI and Agentic AI

- Traditional AI performs predefined tasks based on input, whereas Agentic AI acts autonomously to achieve goals through decision-making and interaction with its environment.

## 1. Traditional AI

- Systems designed to **perform specific tasks** based on predefined rules or trained models.
- Works in a **reactive manner** (responds when given input).

## Agentic AI

- Systems designed as **autonomous agents** that can **plan, decide, and act independently** to achieve goals.

## 2. Control & Autonomy

- **Traditional AI** → Low autonomy
  - Needs human instructions for each task
- **Agentic AI** → High autonomy
  - Can take **initiative and make decisions on its own**

## 3. Behavior

- **Traditional AI**
  - Reactive (input → output)
  - No self-driven actions
- **Agentic AI**
  - Proactive (goal → plan → action → feedback)
  - Continuously improves actions

## 4. Learning & Adaptation

- **Traditional AI**

- Learns during training phase only
- Limited adaptability after deployment

- **Agentic AI**

- Can **learn continuously**
- Adapts based on environment and feedback

## 5. Decision Making

- **Traditional AI**

- Follows fixed logic or trained patterns

- **Agentic AI**

- Makes **dynamic decisions**
- Can evaluate multiple strategies before acting

- **6. Interaction with Environment**

- **Traditional AI**

- Minimal interaction
- Works on static input

- **Agentic AI**

- Constant interaction with environment
- Senses → decides → acts → observes

## **7. Architecture**

- **Traditional AI**

- Model-centric (ML/DL models)

- **Agentic AI**

- Agent-based:
    - Perception
    - Reasoning
    - Planning
    - Action



- **Example**

- **Traditional AI**

- Spam email filter
- Recommendation systems
- Image classifier

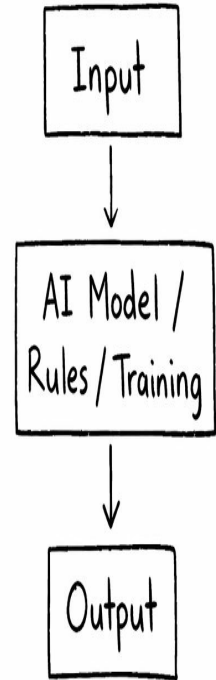
- **Agentic AI**

- Autonomous robots
- Self-driving systems
- Multi-agent systems (buyer-seller agents)

# Differences between Traditional AI and Agentic AI

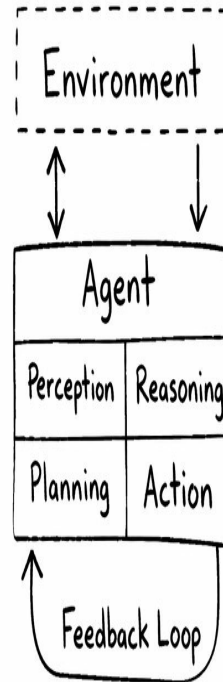
| Feature                | Traditional AI    | Agentic AI                     |
|------------------------|-------------------|--------------------------------|
| <b>Control</b>         | Centralized       | Decentralized                  |
| <b>Behavior</b>        | Predefined rules  | Goal-driven                    |
| <b>Decision Making</b> | Limited           | Autonomous                     |
| <b>Interaction</b>     | Minimal           | High (multi-agent interaction) |
| <b>Flexibility</b>     | Low               | High                           |
| <b>Learning</b>        | Sometimes static  | Adaptive & learning            |
| <b>Example</b>         | Rule-based system | Multi-agent system             |

## Traditional AI



Reactive

## Agentic AI



| Traditional AI | Agentic AI |
|----------------|------------|
|----------------|------------|

|          |           |
|----------|-----------|
| Reactive | Proactive |
|----------|-----------|

|              |               |
|--------------|---------------|
| Low Autonomy | High Autonomy |
|--------------|---------------|

|        |         |
|--------|---------|
| Static | Dynamic |
|--------|---------|

# Generative AI

**Generative AI** is a type of artificial intelligence that can **create new content** such as text, images, code, music, or videos based on the input it receives.

Generative AI is a subset of AI that **learns patterns from existing data and generates new, similar content** in response to user prompts.

1. It is trained on **large datasets** (books, images, code, etc.)
2. It learns **patterns, structure, and relationships**
3. When you give a prompt, it **predicts and generates the most likely output**

Example:

If you type *“Write a story about a robot”*, it generates a completely new story using learned patterns.

# Examples of Generative AI

- **Text generation:** ChatGPT writing answers, essays
- **Image generation:** Tools creating artwork from text
- **Code generation:** AI suggesting or writing programs
- **Music/video generation:** AI composing songs or videos